

第8章 语义分割



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本次课程内容

1. 语义分割概念

5. 注意力机制

2. 基于CNN的语义分割

6. 结合传统模型分割

3. 基于FCN的语义分割

7. 语义分割前沿技术

4. 语义分割网络架构

1. 语义分割概念

(1) 什么语义 (semantic)

- 语义概念的来源： 语言所蕴含的意义就是语义 (semantic)
- 符号是语言的载体，符号本身没有任何意义，语言的含义就是语义。

例如，今天的节目太精彩了！

- ◆ **图像或视频具有语义：**图像或视频是从三维客观世界采集得到的信息，其中**蕴含着现实世界中事物所代表概念的含义**，以及这些含义之间的关系，就是其语义。例如：



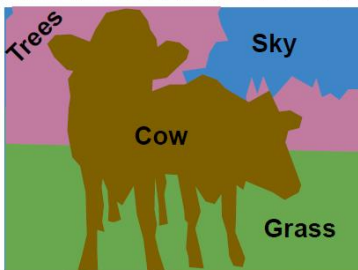
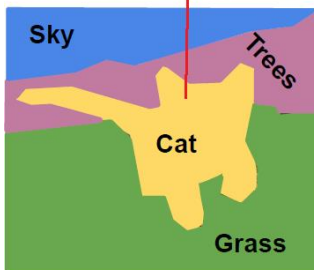
图像语义理解实例

- ◆ 有山吗？ 哪部分是山？
- ◆ 有几棵树？



(2) 图像语义分割概念 (semantic segmentation)

- ◆ 把图像或视频内容按照其含义，进行对象分割
- ◆ 像素级分割本质:标注出像素所属类别 ($\text{height} \times \text{width} \times 1$)



语义分割的本质问题

◆语义图的分辨率应与原始输入图像的分辨率相匹配



Input

segmented

- 1: Person
- 2: Purse
- 3: Plants/Grass
- 4: Sidewalk
- 5: Building/Structures

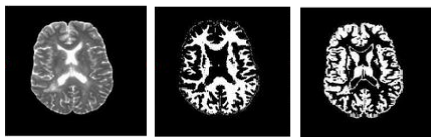
3	3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	1	1	1	1	3	3	3	5	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	5	5	5	5	5	5	5	5
5	5	3	3	3	3	1	1	3	3	5	5	5	5	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	5	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	4	4	5	5	5	5
4	4	4	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4	4
3	3	3	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4	4

Semantic Labels

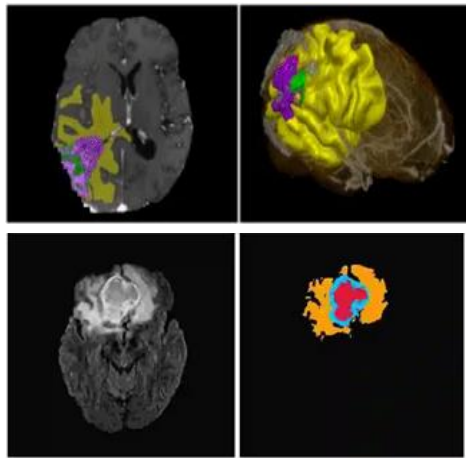
(3) 语义分割的应用



自动驾驶



医学图像分割

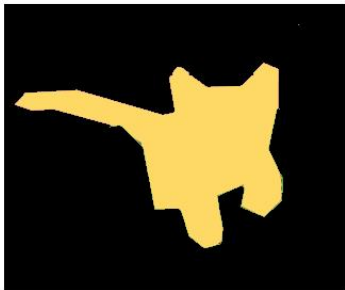
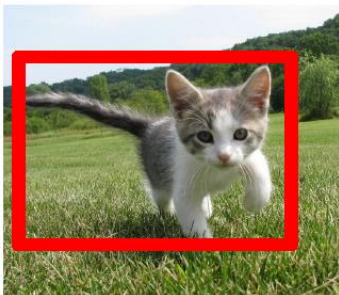


医学影像病灶分割



遥感图像目标识别

(4) 单实例与多实例分割

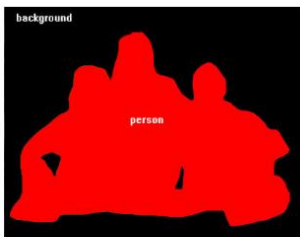


实例分割

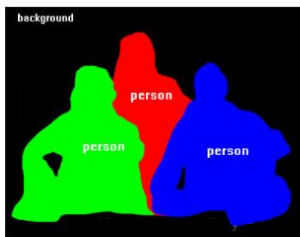
◆ 实例分割：对图像中的每个像素划分出对应的实例类别



输入图像



语义分割结果



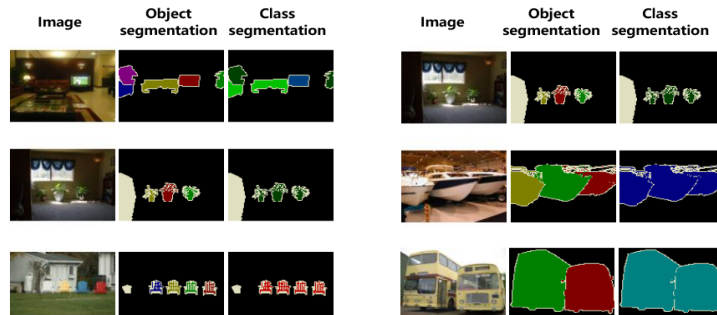
实例分割结果

PASCAL VOC2011 Example Segmentations

Below are training examples for the segmentation taster, each consisting of:

- **the training image**
- **the object segmentation**
pixel indices correspond to the first, second, third object etc.
- **the class segmentation**
pixel indices correspond to classes in alphabetical order (1=aeroplane, 2=bicycle, 3=bird, 4=boat, 5=bottle, 6=bus, 7=car, 8=cat, 9=cow, 10=deer, 11=dog, 12=horse, 13=sheep, 14=sofa, 15=train, 16=tv/monitor)

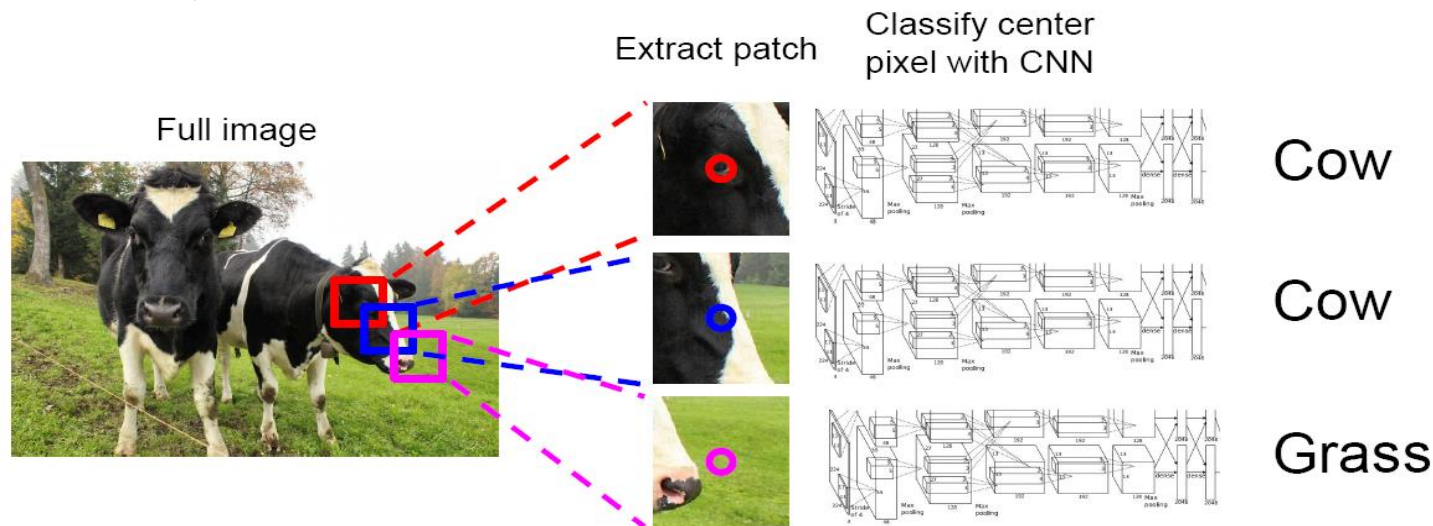
For both types of segmentation image, index 0 corresponds to background and index 255 corresponds to 'void' or unlabelled.



数据集

2. 基于CNN的语义分割

语义分割的滑动窗口的想法，语义分割问题转为图像**patch**的分类问题

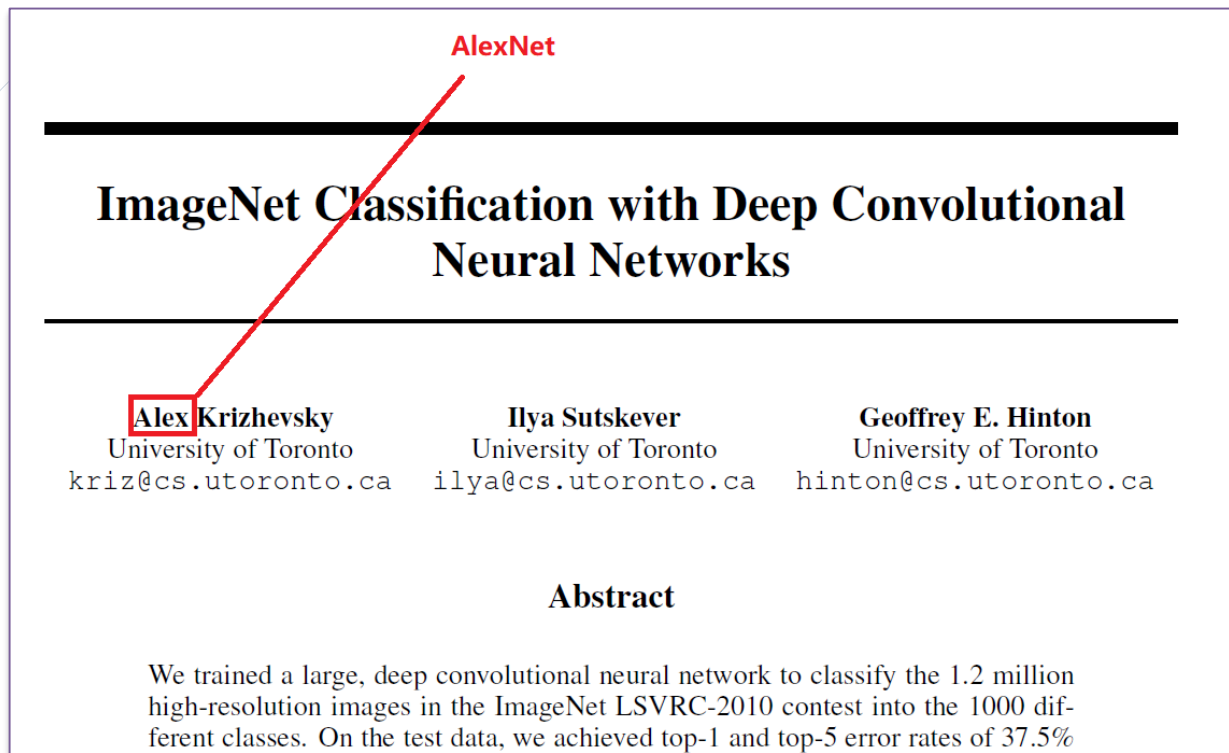


存在的主要问题：计算的复杂度太高，采用CNN进行解决

基于CNN的语义分割

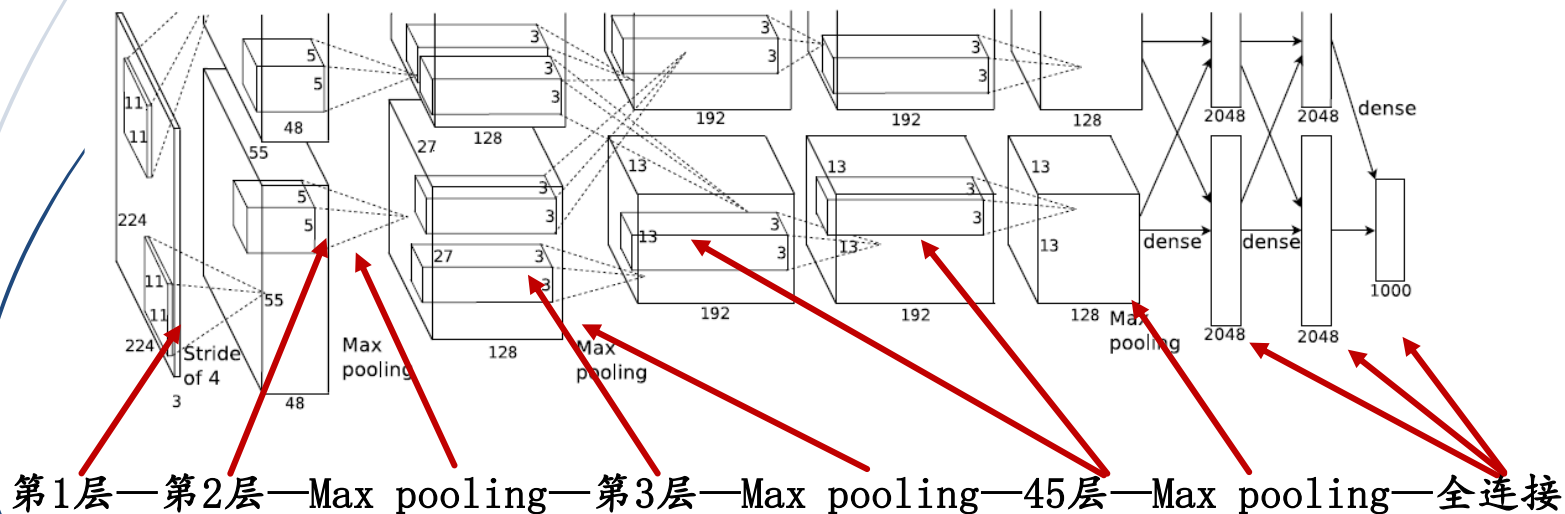
- ◆ 先出现AlexNet
- ◆ 进一步出现FCN的全卷积网络的语义分割算法

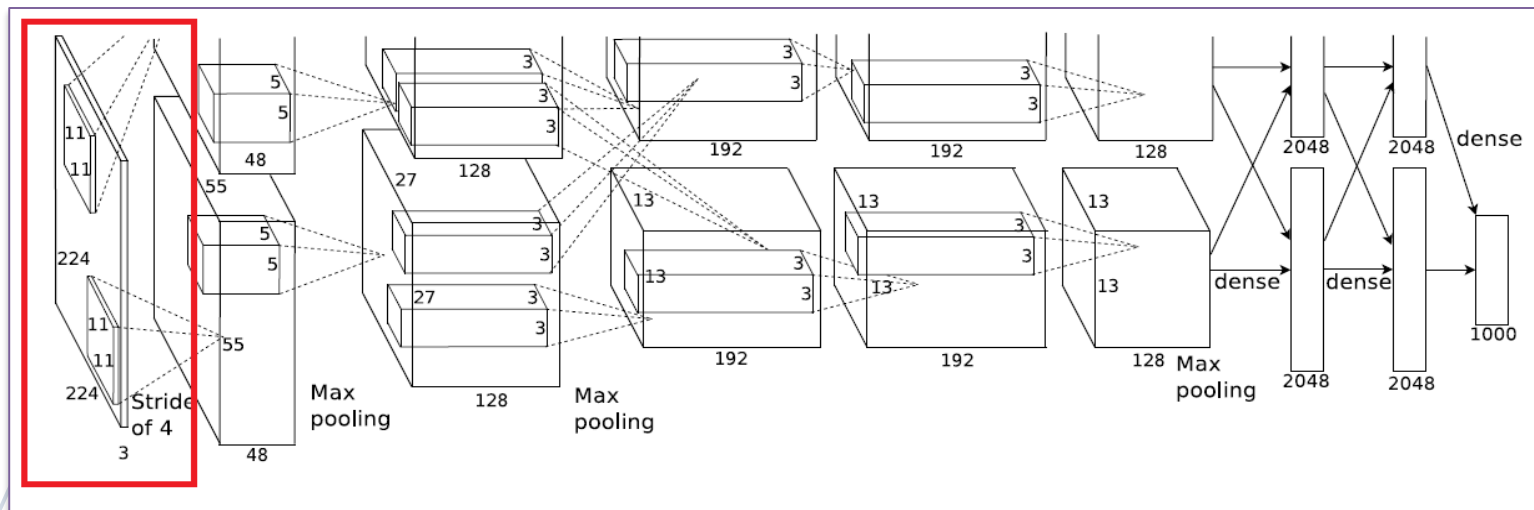
AlexNet介绍——图像分类功能



AlexNet工作的主要原理

- ◆ AlexNet 以准确率84.6%赢得了ILSVRC-2012，由Krizhevsky等人提出。
- ◆ AlexNet包含5个卷积层，3个全连接层，结构如下图所示：





- ◆ 第一层为输入层，就是一幅224X224X3的image， 第一个卷积层， kernel 是11X11X3
- ◆ 每个11X11X3的图像区域卷积操作成为一个新的像素点的一个信道。卷积的步长(stride)为4， 说明224 X 224 被卷积成为是 55X55 的大小

AlexNet的优点

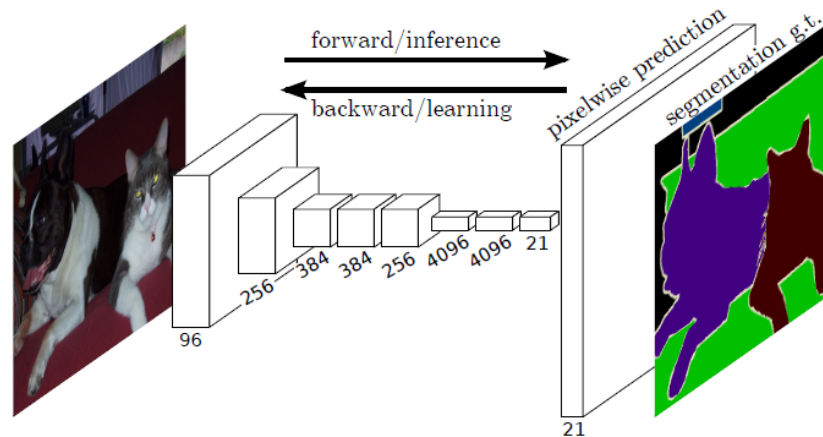
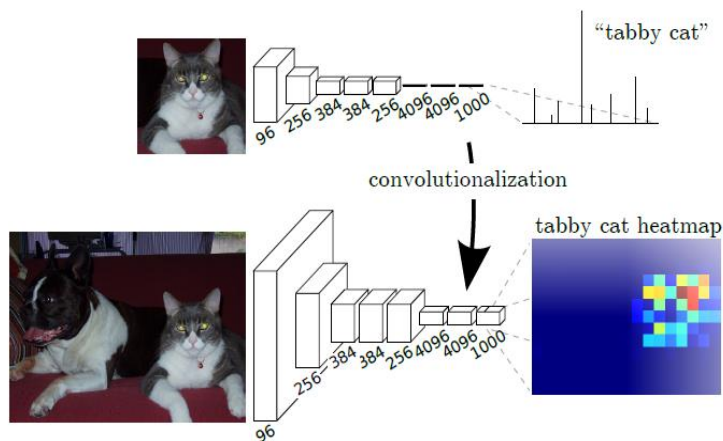
- ◆ 网络结构中五个卷积层和3个全连接层，具有六千万的参数以及65万个神经元；
- ◆ 利用softmax函数实现了1000个类别的分类
- ◆ 利用高效的GPU实现卷积操作
- ◆ 进一步通过该模型的变体在ILSVRC-2012的竞赛中取得了优异的成绩
- ◆ 出现的问题： 采用全链接网络（Full Connected Layers—FCN）进行处理， 要求输入图像分辨率是固定尺度的，改进措施中全链接层换为全卷积层

3. 基于FCN的语义分割

- ◆ 2014年，加州大学伯克利分校的Long等人提出**FCN 分割方法**，在**不带有全连接层的情况下**能进行密集预测。
- ◆ 分割图像可以是任意大小，FCN提高了处理速度
- ◆ FCN为语义分割技术的发展奠定基础

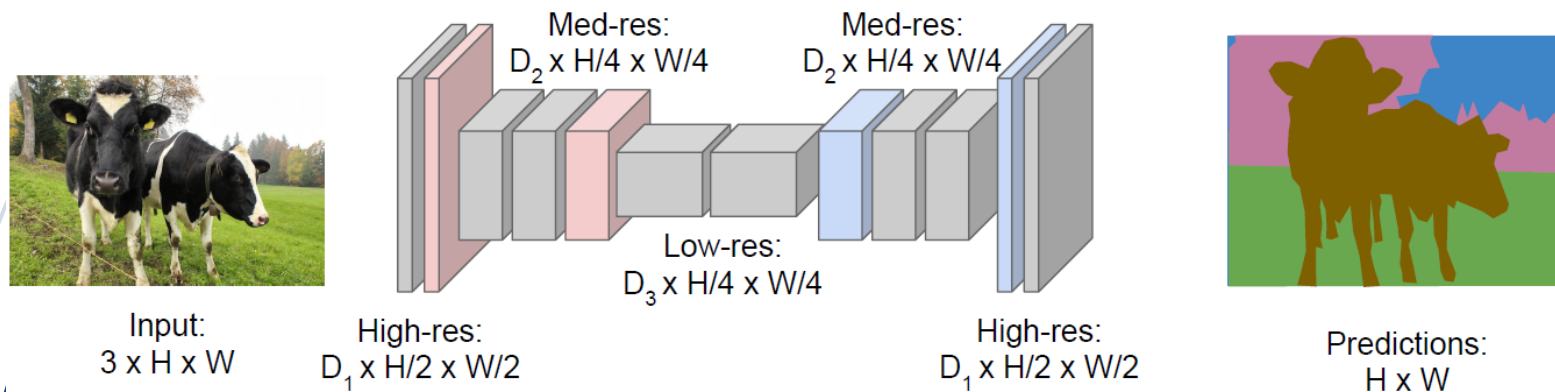
FCN设计的主要思想

- ◆ 将现有的CNN网络（如 AlexNet）作为网络的编码模块，把全连接层转换为卷积层
- ◆ FCN是一种端对端（end-to-end）分割结构，利用标签（ground truth）作监督



FCN设计的主要思想

- ◆ 设计中采用卷积层： 包括下采样 downsampling 和上采样 upsampling



上采样的措施

Nearest Neighbor

1	2
3	4

Input: 2 x 2



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Output: 4 x 4

“Bed of Nails”

1	2
3	4

Input: 2 x 2



1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Output: 4 x 4

上采样的措施

◆ Max Unpooling

Max Pooling

Remember which element was max!

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

Input: 4 x 4

5	6
7	8

Output: 2 x 2

Rest of the network

Max Unpooling

Use positions from pooling layer

1	2
3	4

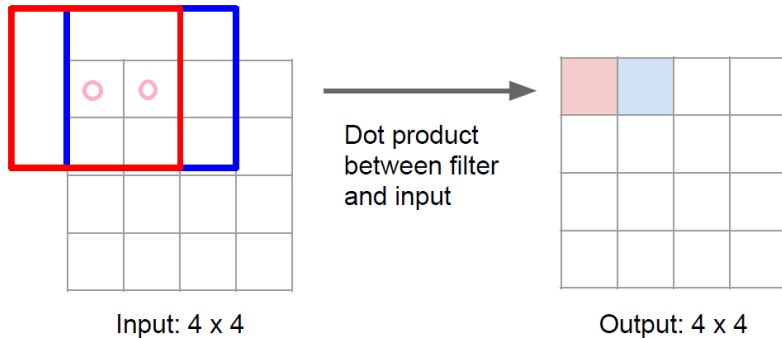
Input: 2 x 2

0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

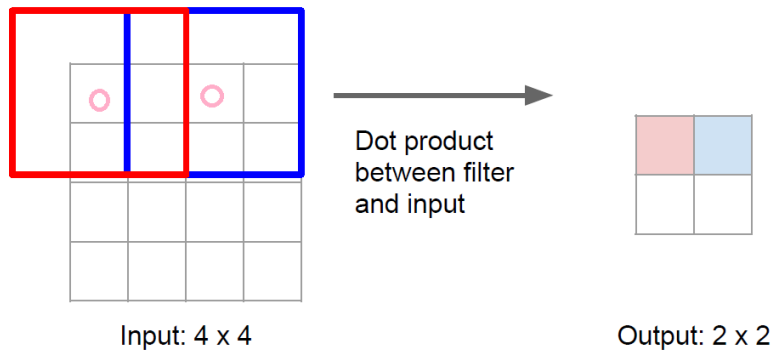
Output: 4 x 4

卷积概念的回顾

◆ Typical 3 x 3 convolution, stride 1 pad 1



◆ 3 x 3 convoluti



FCN的层次卷积结构

- ◆ 对原图像进行卷积conv1、pool1后原图像缩小为1/2
- ◆ 对图像进行第二次conv2、pool2后图像缩小为1/4
- ◆ 进行第三次卷积操作conv3、pool3缩小为原图像的1/8，此时保留pool3的featureMap
- ◆ 再对图像进行第四次卷积操作conv4、pool4，缩小为原图像的1/16，保留pool4的featureMap
- ◆ 最后对图像进行第五次卷积操作conv5、pool5，缩小为原图像的1/32
- ◆ 把的全连接变成卷积操作conv6、conv7，图像featureMap数量改变，图像大小依然为原图的1/32，此时featureMap成为热图heatMap

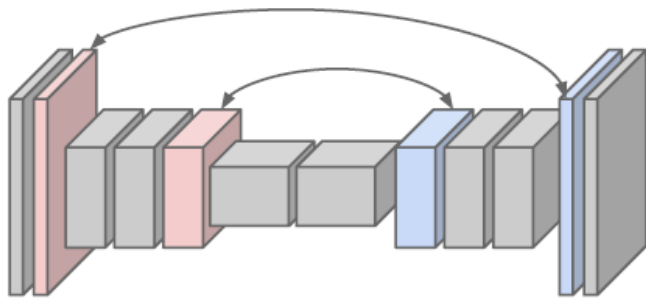
FCN设计中的损失函数

- ◆ 损失函数是最后一层空域中的损失和，可以利用各层梯度损失的和计算求得：

$$\ell(\mathbf{x}; \theta) = \sum_{ij} \ell'(\mathbf{x}_{ij}; \theta)$$

FCN方法优点

- ◆ 利用已经训练好的**supervised pre-training**的网络，无需从头训练，只需要**微调fine-tuning**即可，训练效率高
- ◆ 跨层融合粗细语义，能学习得到较好的语义特征，得到满意的空间预测结果。



FCN语义分割方法缺点

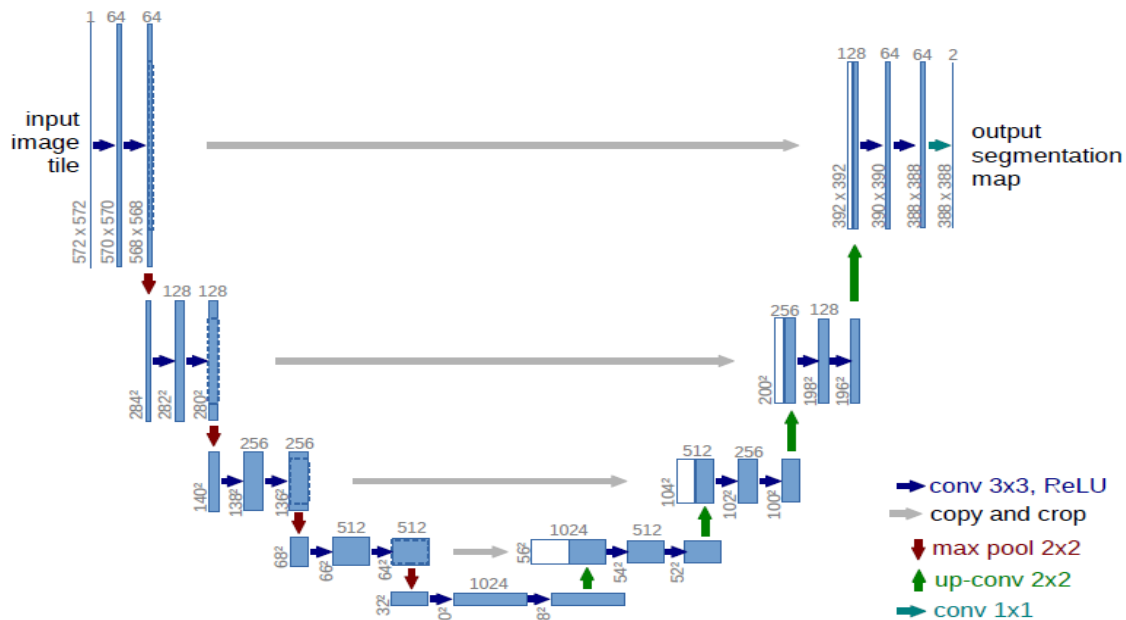
- ◆ 得到的结果不够精细。进行**8倍**上采样虽然比**32倍**的效果好了很多，但是上采样的结果还是比较模糊和平滑，对图像中的细节不敏感。
- ◆ 对像素进行分类，没有充分考虑像素与像素之间的关系

4. 语义分割网络架构

- ◆ 人们提出了不同形式的网络结构解决语义分割问题，例如：
 - 编码器-解码器(**encoder-decoder**)结构
 - 空洞卷积的结构

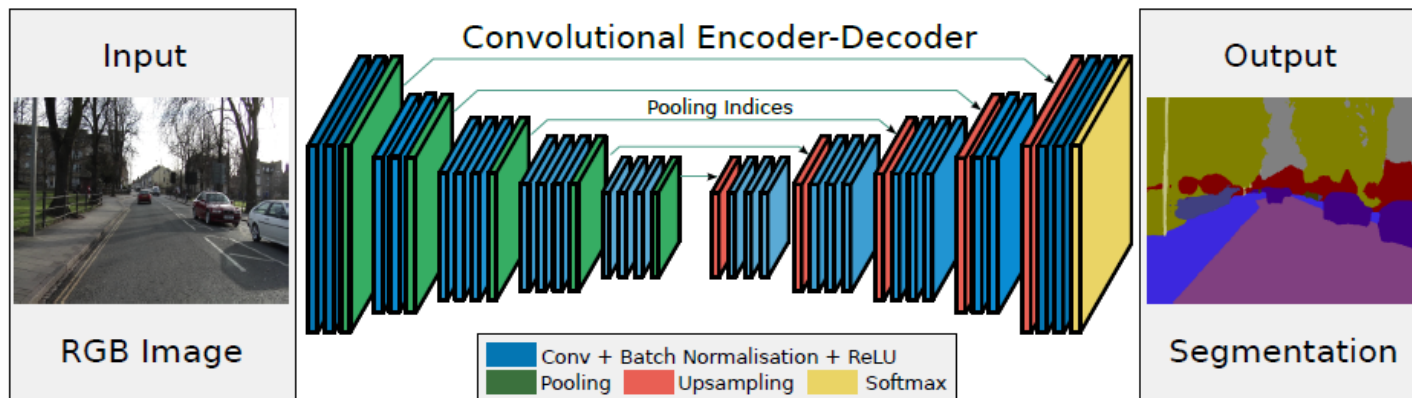
(1) encoder-decoder

◆ 典型结构为U-Net网络



SegNet 结构

◆ SegNet典型的encoder-decoder结构



(2) 空洞卷积的结构

◆ 空洞卷积的结构

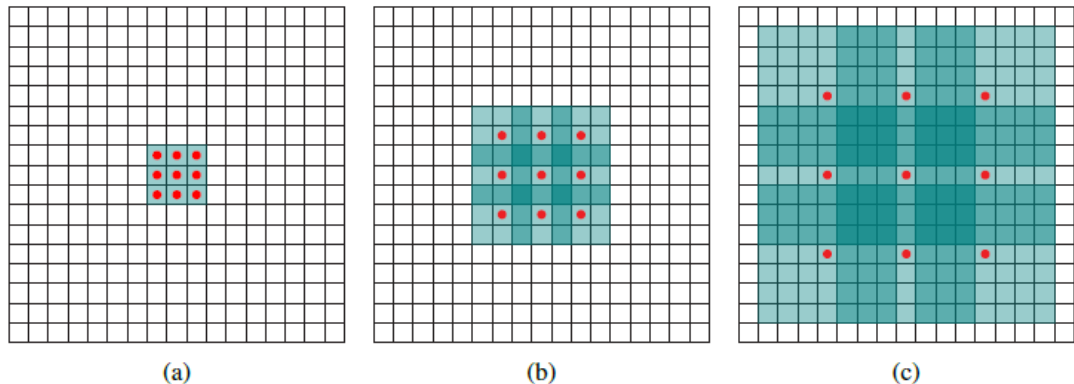


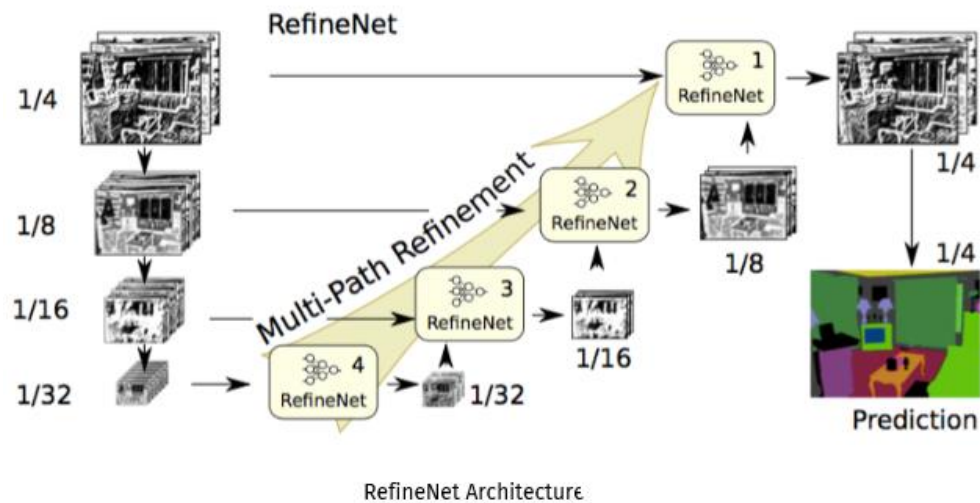
Figure 1: Systematic dilation supports exponential expansion of the receptive field without loss of resolution or coverage. (a) F_1 is produced from F_0 by a 1-dilated convolution; each element in F_1 has a receptive field of 3×3 . (b) F_2 is produced from F_1 by a 2-dilated convolution; each element in F_2 has a receptive field of 7×7 . (c) F_3 is produced from F_2 by a 4-dilated convolution; each element in F_3 has a receptive field of 15×15 . The number of parameters associated with each layer is identical. The receptive field grows exponentially while the number of parameters grows linearly.

DeepLab (v1和v2)

- ◆ 使用了空洞卷积，提出了在空间维度上实现金字塔型的空洞池化方法
- ◆ 使用了全连接条件随机场。
- ◆ 空洞卷积在不增加参数数量的情况下增大了感受野，改善分割分辨率

RefineNet

- ◆ 所有组件遵循残差连接的设计方式



语义分割典型backbone

- ◆ LeNet、ResNet 、DenseNet 、 VGGNet 、 GoogLeNet
- ◆ Res2Net、ResNeXt、InceptionNet
- ◆ SqueezeNet、ShuffleNet、SENet、DPNet
- ◆ MobileNet、NasNet、DetNet
- ◆ EfficientNet等

常结合注意力机制

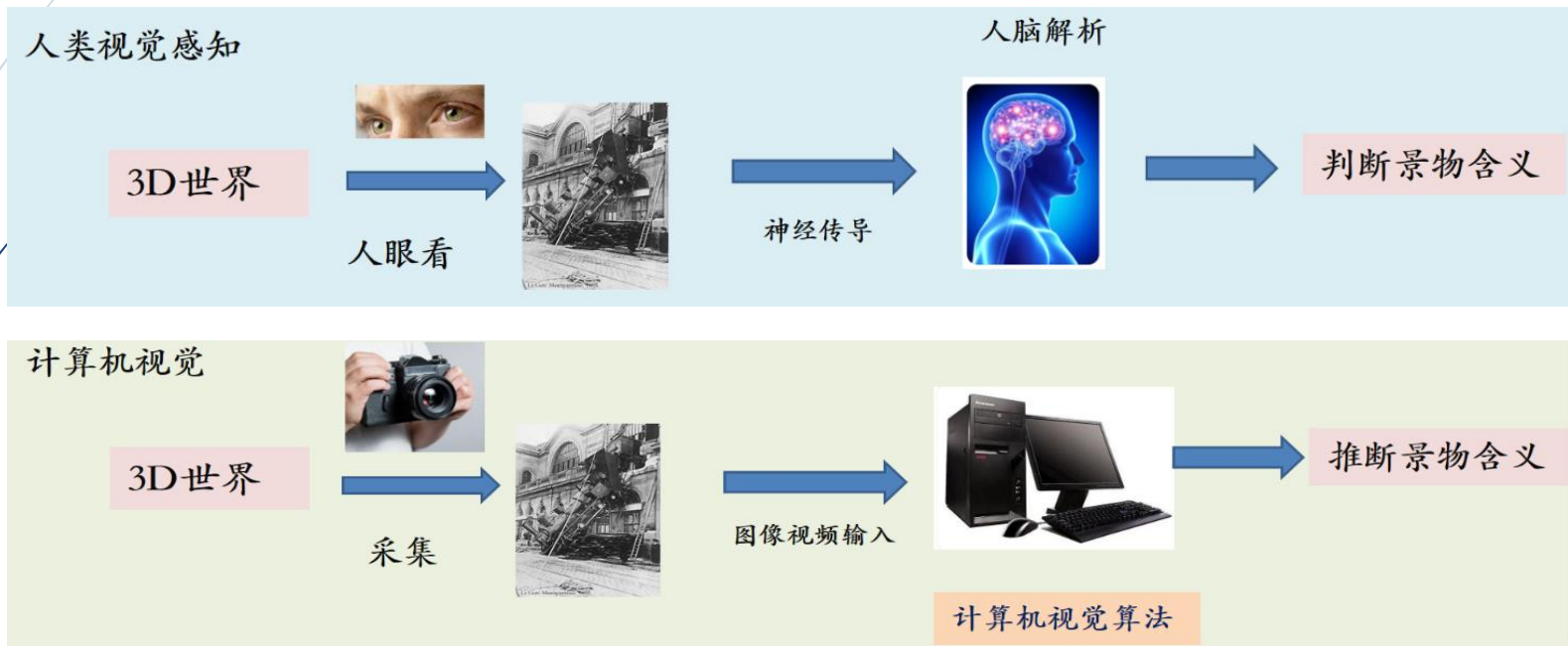
- ◆ SE Net
- ◆ Non-local Neural Network
- ◆ CcNet
- ◆ GC-Net
- ◆ Gate
- ◆ CBAM
- ◆ Dual Attention
- ◆ Spatial Attention
- ◆ Channel Attention

常结合多尺度机制

- ◆ ASPP
- ◆ PPM
- ◆ DCM
- ◆ DenseASPP
- ◆ FPA
- ◆ OCNet
- ◆ MPM

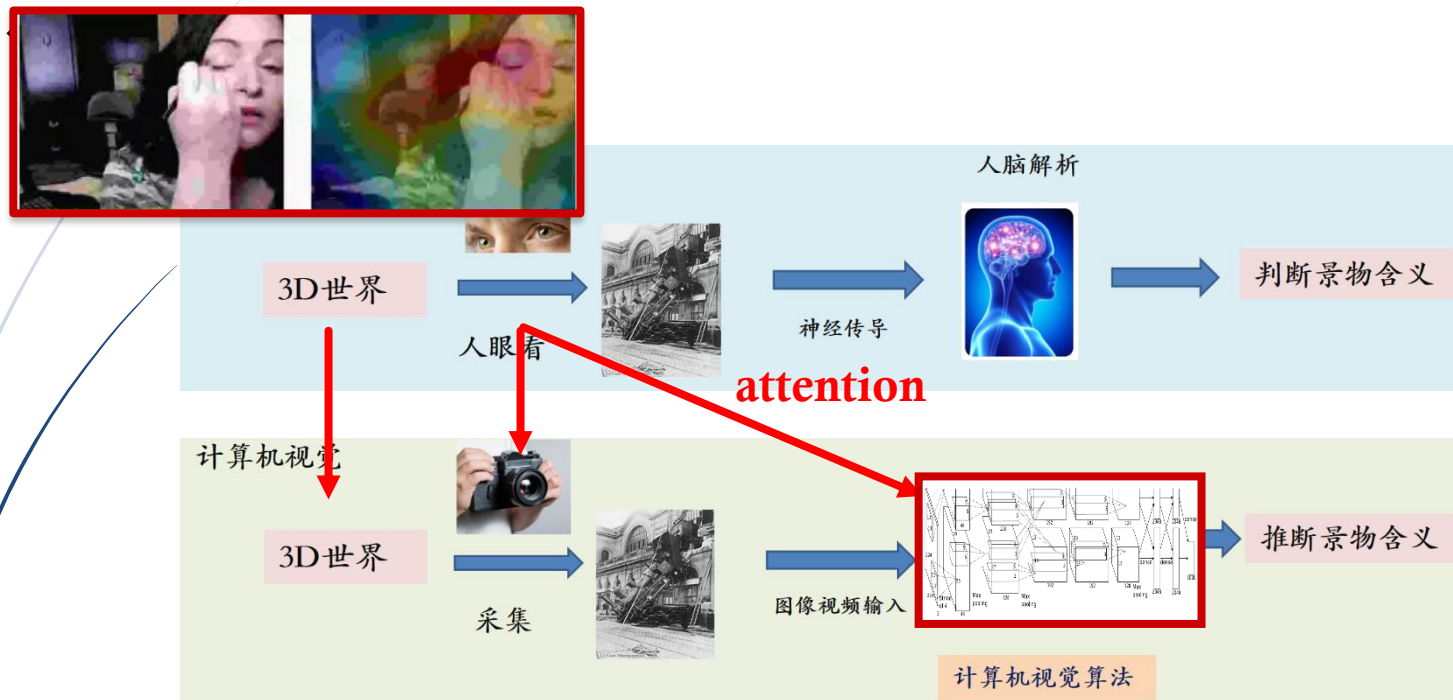
5. 注意力机制

- ◆ 注意力机制提出，为什么需要视觉注意力？
- ◆ 知识回顾—感知概念



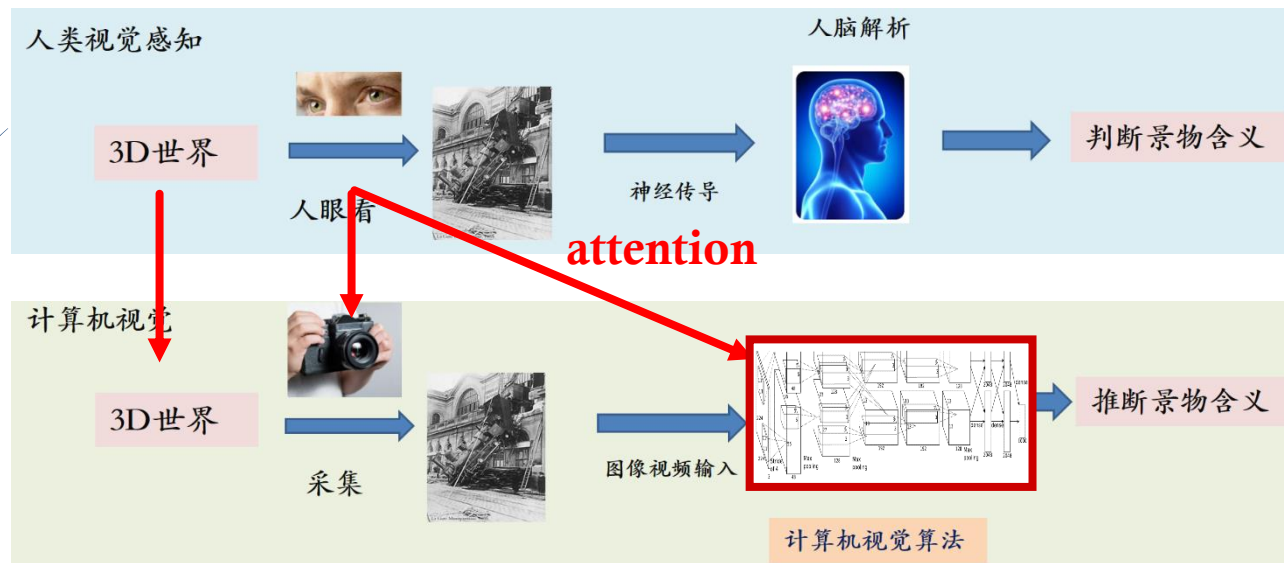
注意力机制引入

- ◆ 注意力机制提出，为什么需要视觉注意力？



注意力机制引入

- ◆ 注意力机制提出，为什么需要视觉注意力？
- ◆ 知识回顾—感知概念



注意力机制引入原因

- ◆ 计算机视觉中的注意力机制（attention）：
 - 让神经网络具有人类注意力感知的能力
 - 让神经网络忽略无关信息
 - 采用“动态加权”的方式使得网络强调部分区域的特征，抑制不相关区域

注意力类型

- ◆ 计算机视觉中的注意力机制 (attention) 常分为：
 - 硬注意力
 - 软注意力
 - 高斯注意力
 - 空间变换
- ◆ **硬注意力 (强注意力)**：就是0/1权重问题，哪些区域是被**注意**，哪些区域**不关注**
- ◆ **软注意力 (Soft-attention)**：在 $[0, 1]$ 间连续分布问题，每个部分被关注的程度高低，用0-1的打分表示。



通道注意力

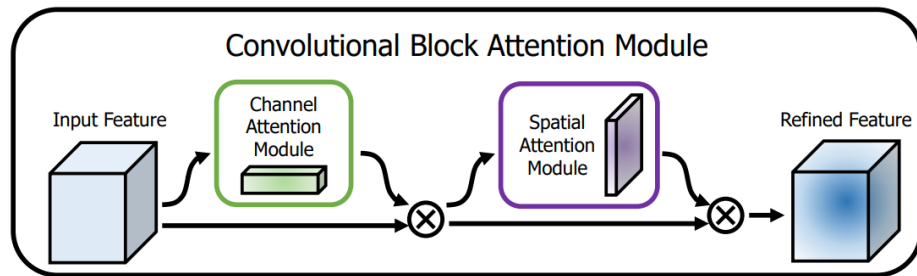
- ◆ **通道注意力**: 通过网络学习的方式来自动获取到每个通道特征的重要程度（权重），最后再为每个通道赋予不同的权重系数
- ◆ 例如:
 - **SE-Net**: 通过注意力来自适应地调整通道之间的特征响应
 - **SK-Net**: 从多尺度特征表征，通过引入多个卷积核，得到每个不同尺度特征图注意力，让网络学习不同尺度特征之间的权重
- ◆ 关键词: Channel-wise attention

例如:

Jiaxing Yan, Hong Zhao, Penghui Bu, Yusheng Jin: Channel-Wise Attention-Based Network for Self-Supervised Monocular Depth Estimation. 3DV 2021: 464-473

空间注意力

- ◆ 空间注意力：通过网络学习的方式来自动获取空间**每个位置的权重掩膜**（mask）并加权输出，从而增强感兴趣的特定目标区域同时弱化不相关的背景区域。
- ◆ 例如：**CBAM**在原有通道注意力的基础上，衔接了一个空间注意力模块（SAM）
 - **Double Attention**：先通道后空间的方式，既要通道注意力，也要空间注意力



自注意力

- ◆ 自注意力：为了减少对外部信息的依赖，尽可能地利用特征内部固有的信息进行注意力的交互。
- ◆ 例如：
 - Transformer架构
 - 何凯明等将自注意力应用到CV领域，并提出了Non-Local模块，通过Self-Attention机制对全局上下文进行建模。

基于attention的语义分割

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segmentation, attention

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- Shanmugasundaram Abinaya, Kandagatla Uttej Kumar, A. Sherly Alphonse:
Cascading Autoencoder With Attention Residual U-Net for Multi-Class Plant Leaf Disease Segmentation and Classification. IEEE Access 11: 98153-98170 (2023)
- Jiawen Cai, Yarong Liu, Pan Qin:
Attention Based Quick Network With Optical Flow Estimation for Semantic Segmentation. IEEE Access 11: 12402-12413 (2023)
- Sabri Deari, Ilkay Öksüz, Sezer Ulukaya:
Block Attention and Switchable Normalization Based Deep Learning Framework for Segmentation of Retinal Vessels. IEEE Access 11: 38263-38274 (2023)
- Hong Huang, Panpan Liu, Jie Liu:
TAGU-Net: Transformer Convolution Hybrid-Based U-Net With Attention Gate for Atypical Meningioma Segmentation. IEEE Access 11: 53207-53223 (2023)
- Yinan Jiang, Chaoliang Zhong, Botao Zhang:
AGD-Linknet: A Road Semantic Segmentation Model for High Resolution Remote Sensing Images Integrating Attention Mechanism, Gated Decoding Block and Dilated Convolution. IEEE Access 11: 22585-22595 (2023)

[+] Refine list

1993

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Guotai Wang (18)
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Guang Yang 0006 (13)
Shaoting Zhang 0001 (13)
Alan L. Yuille (12)
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6,854 more options

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LNCS (204)
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IEEE Access (81)
Comput. Biol. Medicine (64)
Sensors (42)
Biomed. Signal Process. Control. (41)
Remote. Sens. (40)

基于attention的语义分割

- ◆ Mo Zhang, Bin Dong, Quanzheng Li: Joint Attention for Medical Image Segmentation. ISBI 2022: 1–5

6. 结合传统模型分割

- ◆ 区域生长与CNN结合的算法
- ◆ 基于深度学习的蛇模型算法



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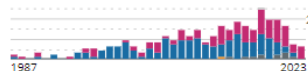
2023

- Wei Wang, Yi Zhang, Gengyu Ge, Qin Jiang, Yang Wang, Lihe Hu:
Indoor Point Cloud Segmentation Using a Modified Region Growing Algorithm and Accurate Normal Estimation. IEEE Access 11: 42510-42520 (2023)
- Jeyshri J., M. Kowsigan:
Adaptive fuzzy-region growing fusion and improved CNN-ANFIS-based automated segmentation and classification of cervical cancer. Concurr. Comput. Pract. Exp. 35(3) (2023)
- Hassan Ali Khan, Xueqing Gong, Fenglin Bi, Rashid Ali:
Novel Light Convolutional Neural Network for COVID Detection with Watershed Based Region Growing Segmentation. J. Imaging 9(2): 42 (2023)
- Priti Bansal, Abhishek Singhal, Kshitiz Gehlot:
Osteosarcoma Detection from Whole Slide Images Using Multi-Feature Non-Seed-Based Region Growing Segmentation and Feature Extraction. Neural Process. Lett. 55(4): 3671-3693 (2023)
- K. Vijila Rani, G. Sumathy, L. K. Shoba, P. Josephin Shermila, M. Eugene Prince:
Radon transform-based improved single seeded region growing segmentation for lung cancer detection using AMPWSVM classification approach. Signal Image Video Process. 17(8): 4571-4580 (2023)
- Ke Liu, Hongchao Ma, Liang Zhang, Xiaoli Liang, Dachang Chen, Yihang Liu:
Roof Segmentation From Airborne LiDAR Using Octree-Based Hybrid Region Growing and Boundary Neighborhood Verification Voting. IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 16: 2134-2146 (2023)

2022

- J. Dafni Rose, K. Vijayakumar, Laxman Singh, Sudhir Kumar Sharma:
Computer-aided diagnosis for breast cancer detection and classification using optimal region growing segmentation with MobileNet model. Concurr. Eng. Res. Appl. 30(2): 181-188 (2022)

[-] Refine list



refine by author

Mohamed Ali Mahjoub (4)
Mahaman Sani Chaibou (4)
David A. Clausi (4)
Bassel Solaiman (4)
Karim Kalti (4)
Raul Queiroz Feitosa (4)
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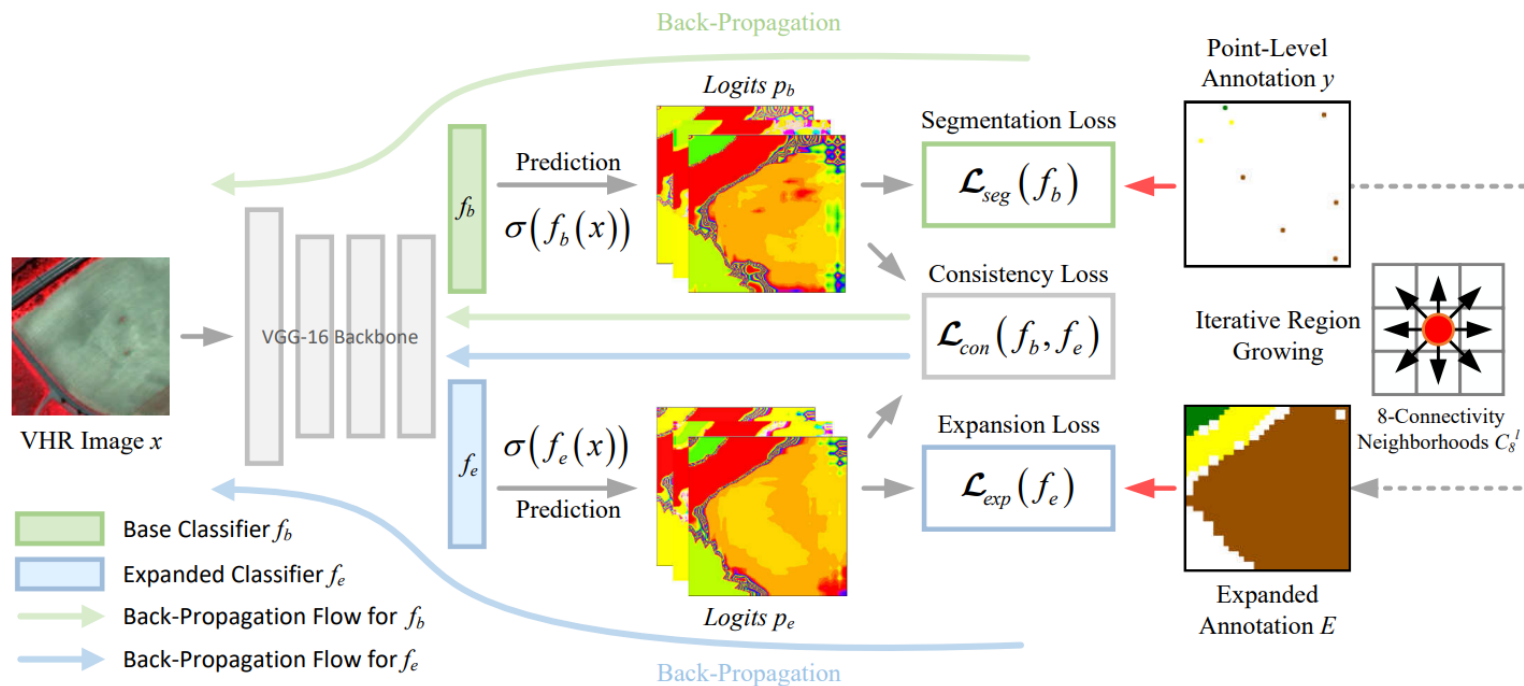
区域生长与CNN结合的算法——实例 Region-Growing Network

Consistency-Regularized Region-Growing Network for Semantic Segmentation of Urban Scenes with Point-Level Annotations

Yonghao Xu, *Member, IEEE*, and Pedram Ghamisi, *Senior Member, IEEE*

- ◆ CRGNet使用两个分类器，包括基本分类器和扩展分类器。
- ◆ 两个分类器共享同一主干网络。
- ◆ 在训练阶段，基础分类器由原始稀疏标注进行监督
- ◆ 扩展分类器则利用区域增长机制从基础分类器生成的扩展标注中学习
- ◆ 两个分类器的预测之间的差异作为损失化。

Region-Growing Network 框架图

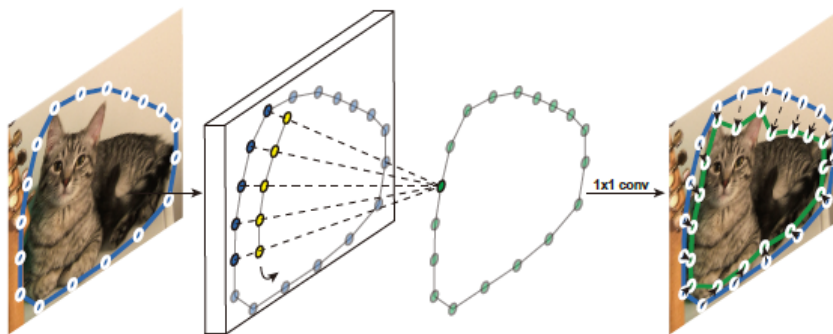


基于深度学习的蛇模型算法

Deep Snake for Real-Time Instance Segmentation

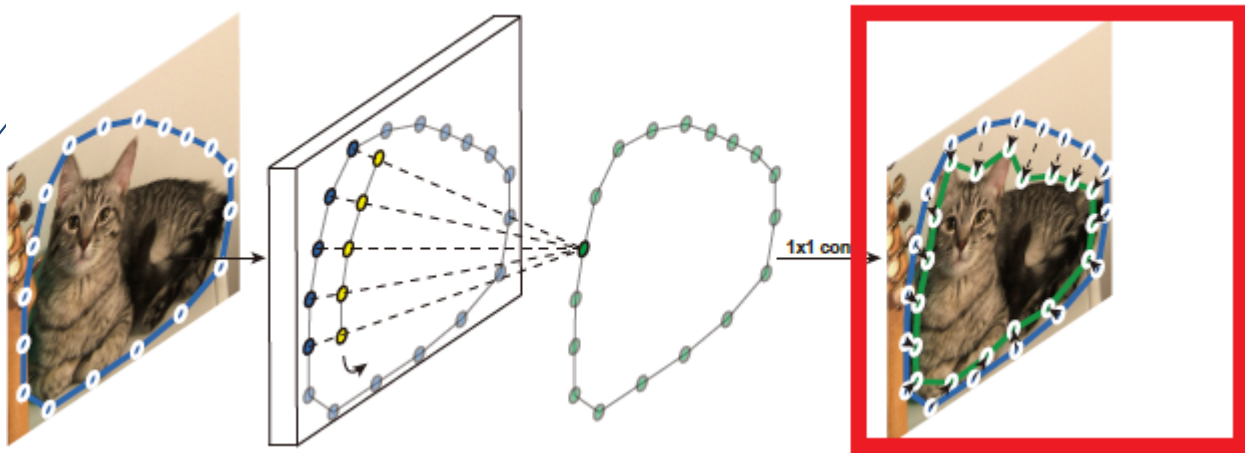
Sida Peng¹ Wen Jiang¹ Huaijin Pi¹ Xiuli Li² Hujun Bao¹ Xiaowei Zhou^{1*}

¹Zhejiang University ²Deepwise AI Lab



基于深度学习的蛇模型算法

- ◆ 使用神经网络将初始轮廓迭代变形为对象边界
- ◆ 初始轮廓作为输入，通过回归得到顶点偏移量位移
- ◆ 根据当前轮廓预测N个偏移，然后将顶点偏移量添加到其顶点坐标来使该轮廓变形。



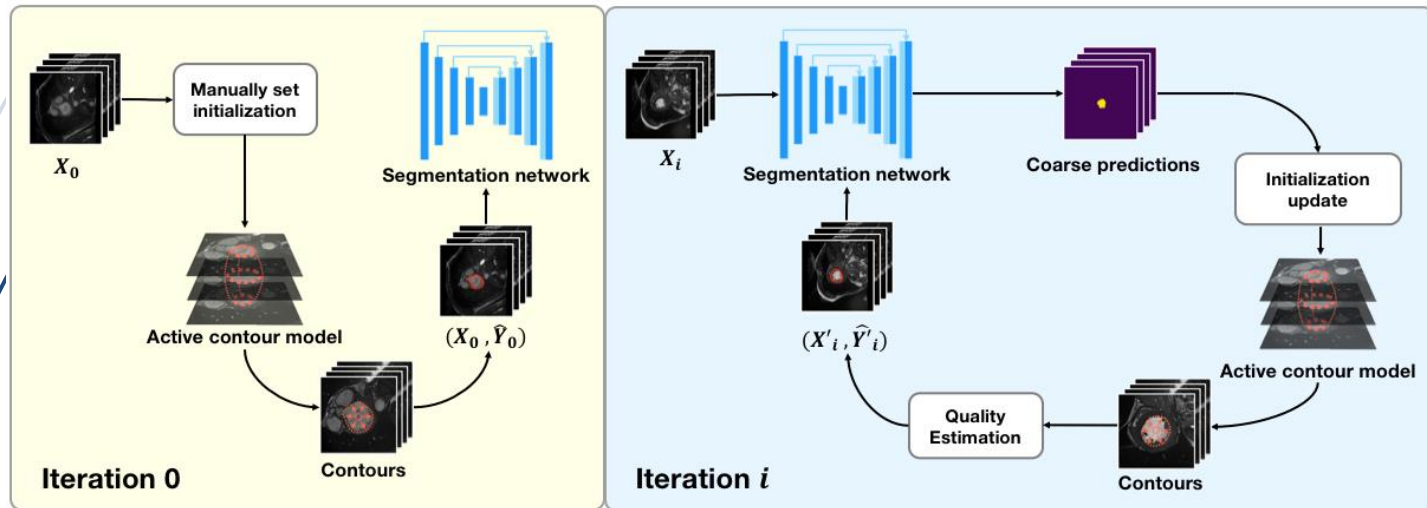
主动轮廓模型 Active Contour Models

- ◆ 主动轮廓线模型：用户在感兴趣目标周围设置初始轮廓线
- ◆ 在内部能量（内力）和外部能量（外力）的作用下变形：外部能量吸引活动轮廓朝物体边缘运动，而内部能量保持活动轮廓的光滑性和拓扑性，当能量达到最小时，活动轮廓收敛到所要检测的物体边缘。
- ◆ 1987年Kass提出Snakes模型以来，各种基于主动轮廓线的图像分割方法迅速发展起来

Active Contour Models和CNN结合的实例

UNSUPERVISED SEGMENTATION FRAMEWORK WITH ACTIVE CONTOUR MODELS
FOR CINE CARDIAC MRI

Du Lianyu* Hu Liwei† Zhang Xiaoyun* Zhong Yumin† Zhang Ya* Wang Yanfeng*



7. 语义分割前沿技术

- ◆ 半监督策略
- ◆ 小样本问题
- ◆ 隐神经表达问题

(1) 半监督策略

- ◆ 半监督学习(**Semi-Supervised Learning, SSL**)是机器学习中的一种学习方法
- ◆ 基于监督学习与无监督学习相结合
- ◆ 半监督学习使用大量的未标记数据, 这样, 对于打标签费时费力可以避免
- ◆ 半监督学习越来越受到人们重视

半监督学习应用

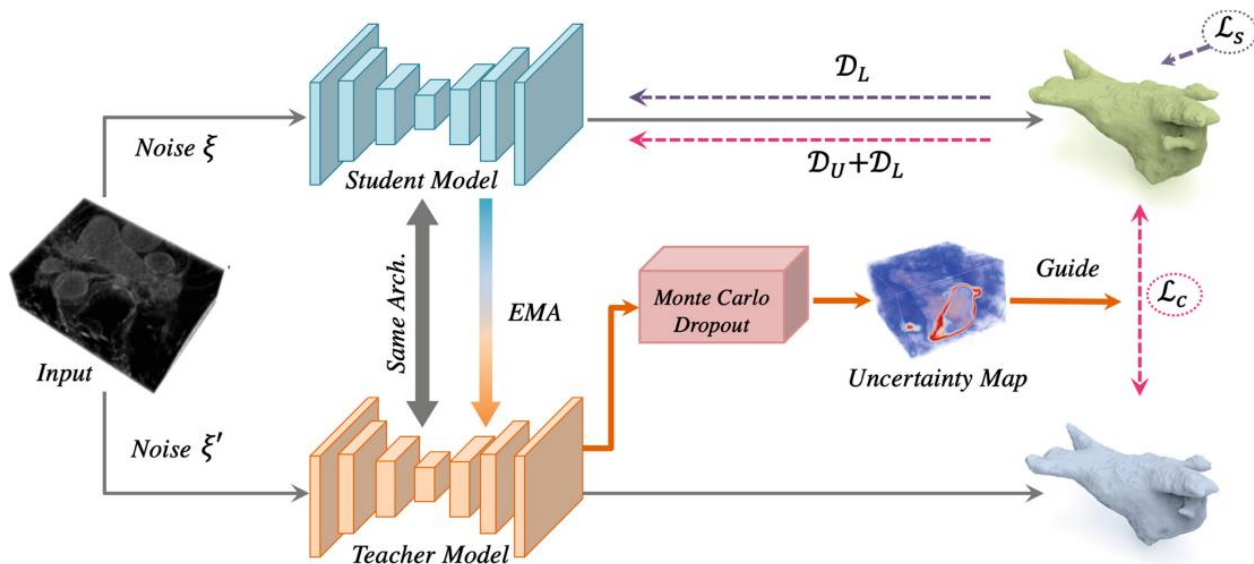
广泛应用：

- 语义分割
- 图像分类
- 三维重建
- 等

实例：语义分割

Uncertainty-Aware Self-ensembling Model for Semi-supervised 3D Left Atrium Segmentation

Lequan Yu^{1(✉)}, Shujun Wang¹, Xiaomeng Li¹, Chi-Wing Fu¹,
and Pheng-Ann Heng^{1,2}



实例：语义分割

Semi-supervised Medical Image Segmentation through Dual-task Consistency

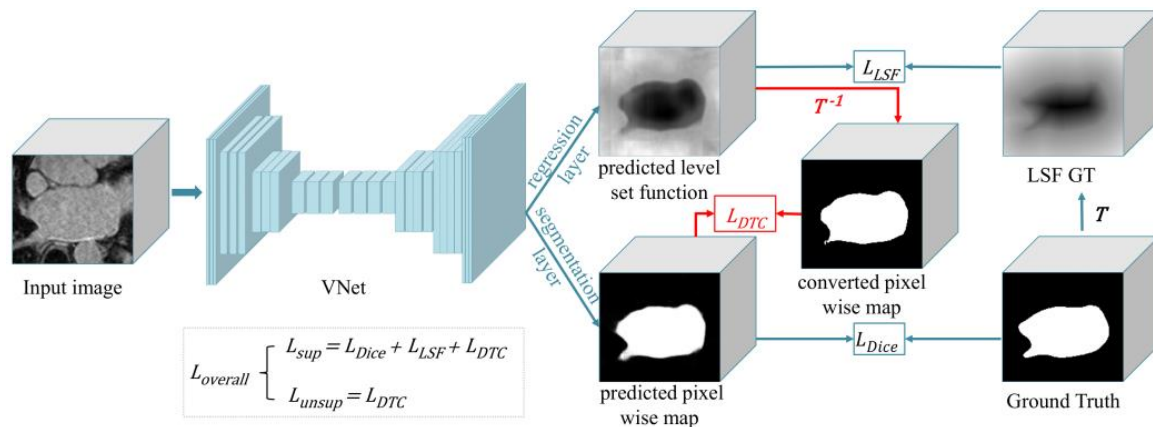
Xiangde Luo^{1,2}, Jieneng Chen³, Tao Song², Guotai Wang^{1*}

¹University of Electronic Science and Technology of China, Chengdu, China

²SenseTime Research, Shanghai, China

³Tongji University, Shanghai, China

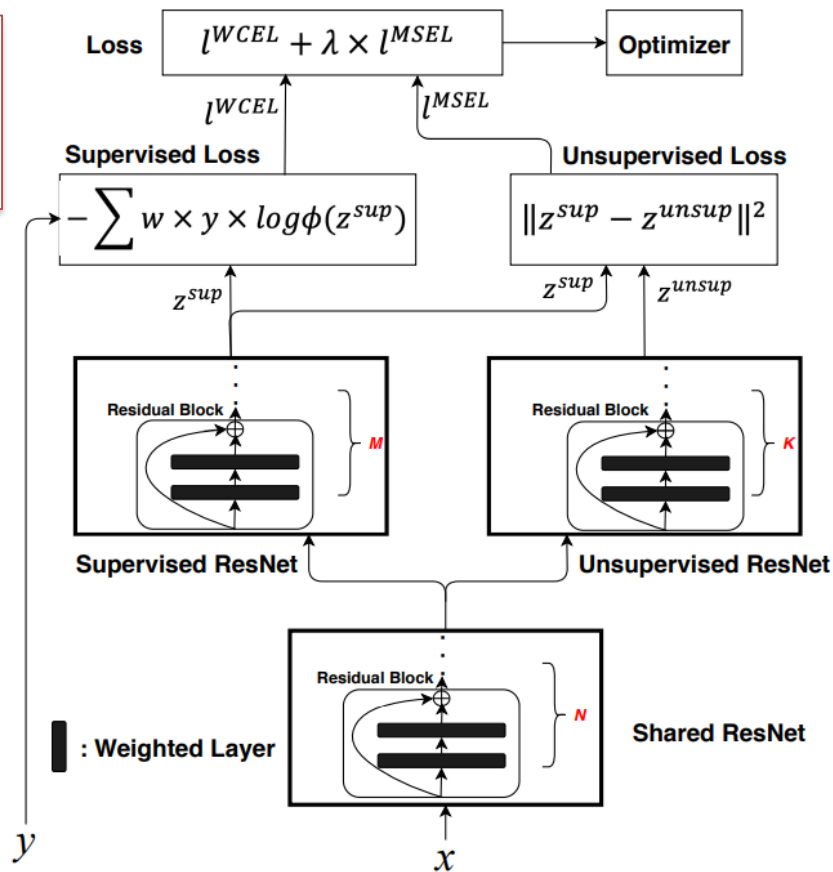
xiangde.luo@std.uestc.edu.cn, chenjn@tongji.edu.cn, songtao@sensetime.com, guotai.wang@uestc.edu.cn



实例：语义分割

Semi-supervised Learning for COVID-19 Image Classification via ResNet

Lucy Nwosu, Xiangfang Li, Lijun Qian, *Senior Member, IEEE*, Seungchan Kim, *Member, IEEE*, Xishuang Dong, *Member, IEEE*



实例：语义分割

Semi-Supervised Medical Image Segmentation via Cross Teaching between CNN and Transformer

Xiangde Luo^{1,2}
Minhao Hu³
Tao Song³
Guotai Wang^{1,2,*}
Shaoting Zhang^{1,2,3}

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¹ University of Electronic Science and Technology of China, Chengdu, China

² Shanghai AI Lab, Shanghai, China; ³ SenseTime, Shanghai, China; * Corresponding author

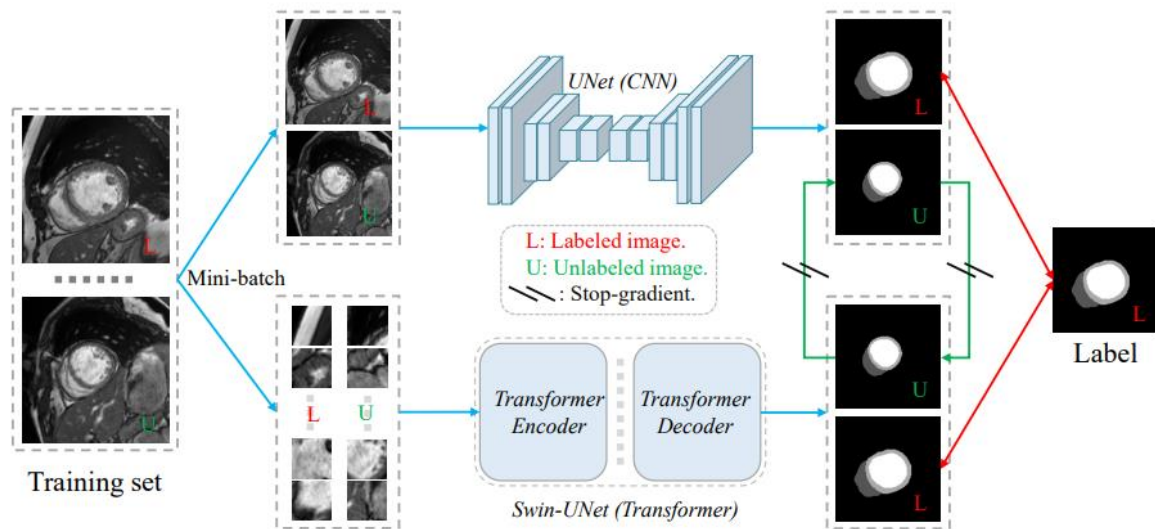


Figure 1: Overview of Cross Teaching between CNN and Transformer.

(2) 小样本学习

- ◆ One shot
- ◆ Few shot
- ◆ 原型学习

One Shot Learning

- ◆ 每个类别需要很少训练样本可以让模型训练达到可以接受的能力。
- ◆ 解决深度学习对数据集要求高的问题

实例：语义分割

- ◆ SG-One: Similarity Guidance Network for One-Shot Semantic Segmentation
- ◆ 关于one-shot语义分割的文章，作者通过输入图像标签对（img-mask）和待分割图像完成分割。

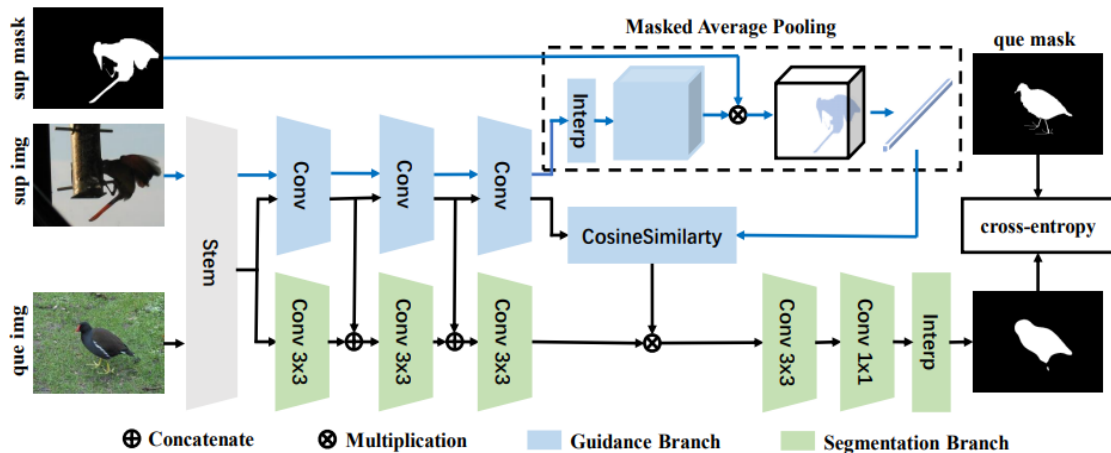
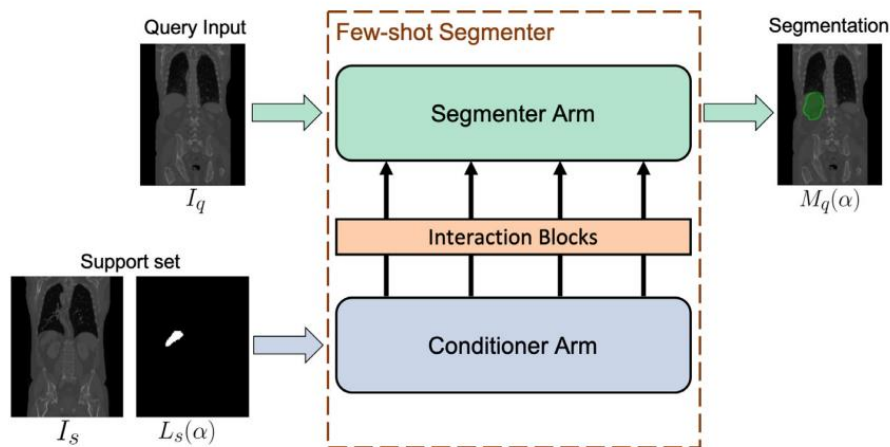


Fig. 2. The network of the proposed SG-One approach. A query image and a labeled support image are fed into the network. Guidance Branch is to extract the representative vector of the target object in the support image. Segmentation Branch is to predict the segmentation masks of the query image. We calculate the cosine distance between the vector and the intermediate features of the query image. The CosineSimilarity maps are then employed to guide the segmentation process. The blue arrows indicate data streams of support images, while the black are for query images. Stem is the conv1 to conv3 of VGG16. Interp refers to the bilinear interpolation operation. Conv is a convolutional block. Conv $k \times k$ is the convolutional filter with a kernel size of $k \times k$.

实例：语义分割

- ◆ 通过输入带标签的图像对和待分割的图像进行处理，通过interaction blocks对两个输入进行交互。



‘Squeeze & excite’ guided few-shot segmentation of volumetric images

Abhijit Guha Roy^{a,b,1,*}, Shayan Siddiqui^{a,b,1}, Sebastian Pölsterl^{a,1}, Nassir Navab^{b,c},
Christian Wachinger^a

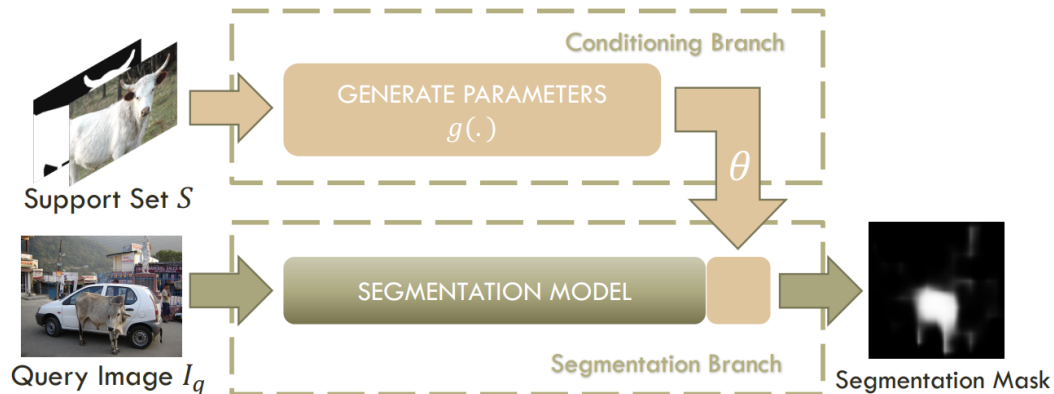
实例：语义分割

◆ 最先提出的工作

One-Shot Learning for Semantic Segmentation

Amirreza Shaban
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Shray Bansal
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Georgia Institute of Technology,
Georgia, USA



原型学习 prototype learning

Few-Shot Semantic Segmentation with Prototype Learning

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Petuum, Inc.
Pittsburgh, PA 15222, USA
Petuum, Inc.
Pittsburgh, PA 15222, USA

Eric P. Xing
eric.xing@petuum.com

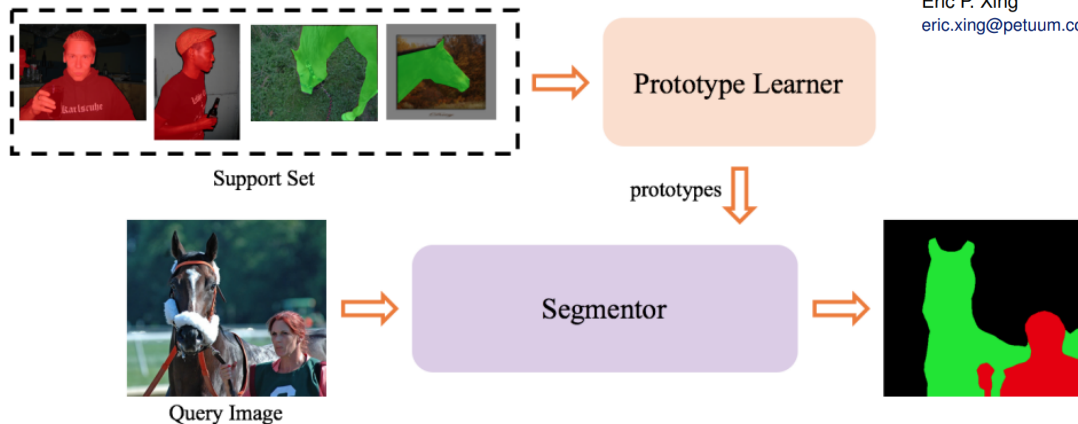


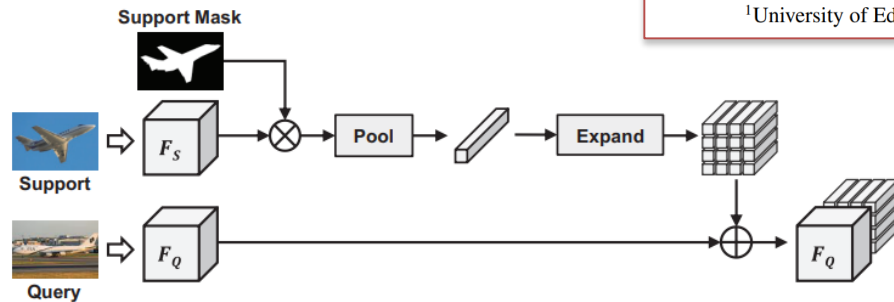
Figure 1: Illustration of a 2-way 2-shot semantic segmentation task. The prototype learner learns the prototypes from the support set and outputs the prototypes to the segmentor. The segmentor takes the query image and the prototypes to predict the segmentation mask. The 1-shot task can be easily extended to k -shot tasks by having k examples for each class in the support set.

prototype learning

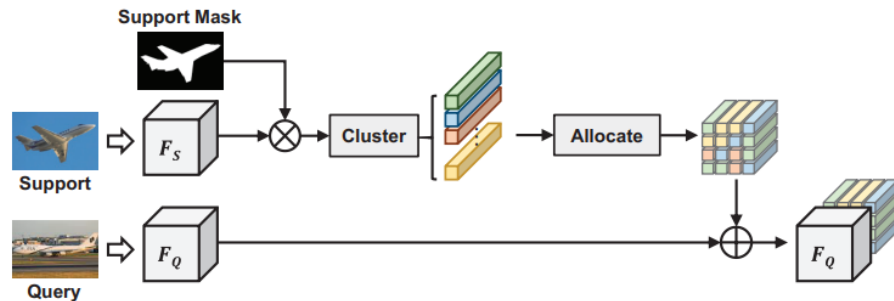
Adaptive Prototype Learning and Allocation for Few-Shot Segmentation

Gen Li^{1,3}, Varun Jampani², Laura Sevilla-Lara¹, Deqing Sun², Jonghyun Kim³, Joongkyu Kim^{3*}

¹University of Edinburgh ²Google Research ³Sungkyunkwan University



(a) Single prototype learning



(b) Adaptive prototype learning and allocation

Figure 1. Comparison between (a) single prototype learning and (b) proposed adaptive prototype learning and allocation. We utilize superpixel-guided clustering to generate multiple prototypes and then allocate them pixel-wise to query feature.

prototype learning

Dynamic Prototype Convolution Network for Few-Shot Semantic Segmentation

Jie Liu^{1*}, Yanqi Bao^{2*}, Guo-Sen Xie^{3,4†}, Huan Xiong⁴, Jan-Jakob Sonke⁵, Efstratios Gavves¹

¹University of Amsterdam, Netherlands ²Northeastern University, China ³The Netherlands Cancer Institute, Netherlands

³Nanjing University of Science and Technology, China ⁴Mohamed bin Zayed University of Artificial Intelligence, UAE

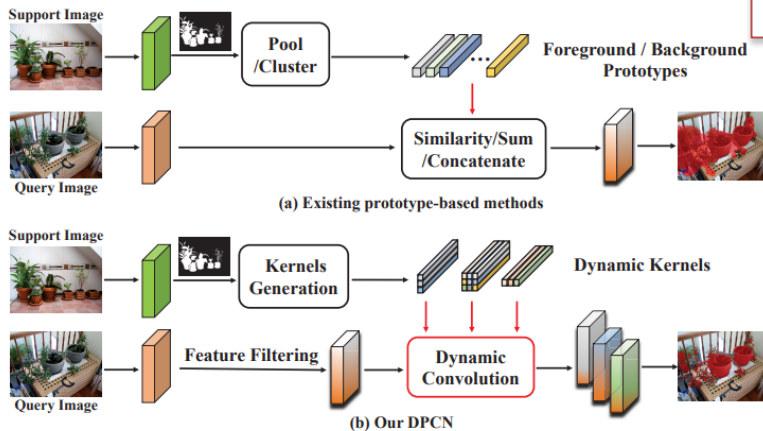
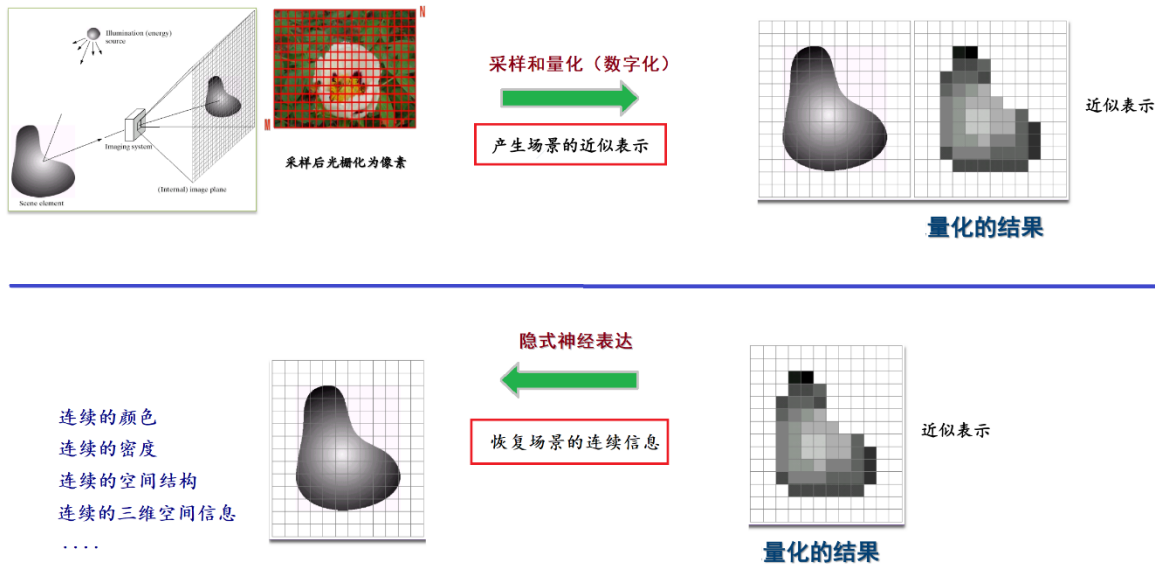


Figure 1. **Comparison between (a) existing prototype-based methods and (b) proposed DPCN.** (a) Existing methods usually adopt mask average pooling or clustering on support foreground to get multiple foreground/background prototypes. Then the information interaction between support and query can be plain operations, e.g., cosine similarity, element-wise sum, and channel-wise concatenation. However, this paradigm fails to segment the intrinsic details of *plants*, because this insufficient interaction cannot well address the appearance and shape variations (e.g., holes and slots in the *plants*) in FSS. (b) **Our DPCN** can well segment *plants* and capture the intrinsic subtle details. This benefits from dynamic convolution on query features with dynamic kernels generated from the support foreground features.

(3) 隐神经表达问题 implicit neural representation

- ◆ 隐式神经表达：应用计算机采样的得到的离散化数据建模，恢复连续空间的信息
- ◆ 该技术研究的意义



隐式神经表达问题 implicit neural representation

◆ 实例：

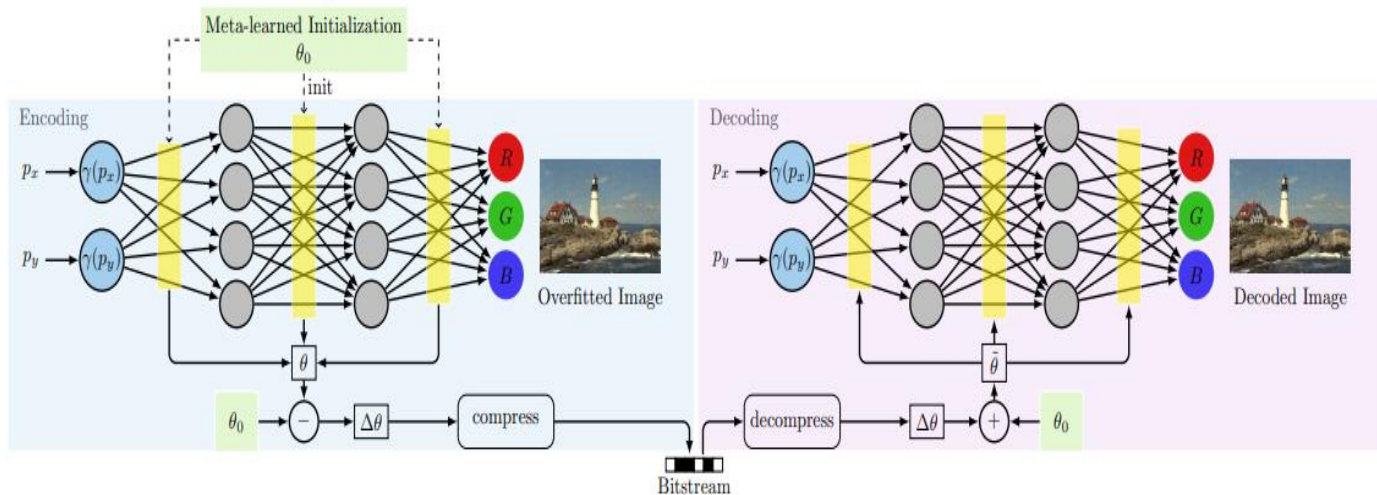
Implicit Neural Representations for Image Compression

Yannick Strümler^{*1}, Janis Postels^{*1}, Ren Yang¹, Luc Van Gool¹, and Federico Tombari^{2,3}

¹ ETH Zurich

² Technical University of Munich

³ Google



隐神经表达

Implicit U-Net for Volumetric Medical Image Segmentation

Sergio Naval Marimont¹ and Giacomo Tarroni^{1,2}

¹ CitAI Research Centre, City, University of London, London, UK
{sergio.naval-marimont, giacomo.tarroni}@city.ac.uk

² BioMedIA, Imperial College, London, UK

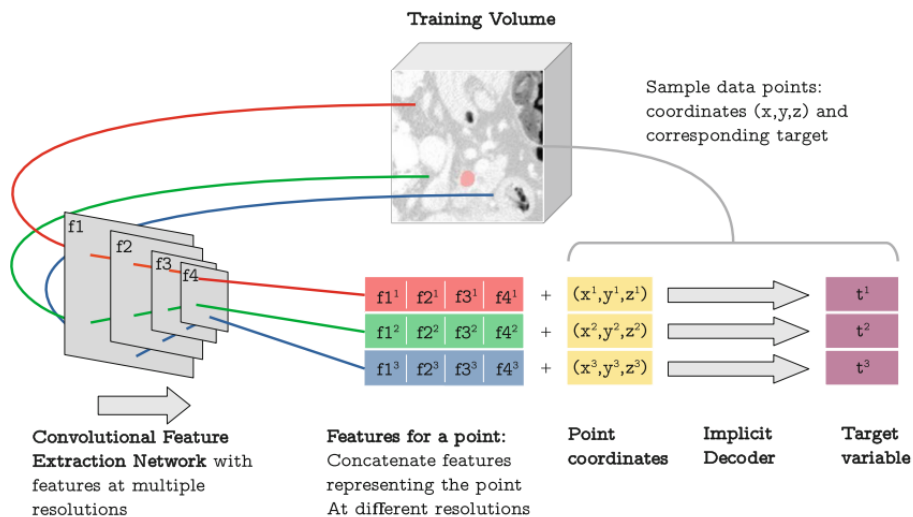


Fig. 1. Diagram of the gather feature vector operation. For example, for $\mathbf{p}$ with coordinates $(8, 16)$, in the 2nd feature map get $(4, 8)$, in 3rd feature map get $(2, 4)$, etc. Features extracted at different resolutions are concatenated with the spatial coordinates and passed to the Implicit decoder.